

Molecular-Driven Multi-View Hypergraph Contrastive Learning for Drug-Drug Interaction Prediction

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Abstract—Recent concerns have arisen over adverse reactions caused by drug combinations, and drug-drug interaction (DDI) prediction helps identify potential risks by forecasting interactions between drugs. Previous methods have primarily explored drug interactions from the superficial level of drug molecules, often overlooking the internal structural information of the molecules. To this end, we propose Mol-HCL, a multi-view hypergraph contrastive learning framework based on molecular view. In this framework, we construct the molecular view to learn the internal information of drug molecules and, based on this, develop structural view and semantic view. These three views collaboratively learn both intra-molecular and inter-molecular information. Subsequently, we incorporate hypernodes into the structural view and design a novel hyperchain, integrating it into the semantic view to capture latent neighbor drug node structural relationships and long-range DDI chain semantic information. After that, contrastive learning is performed between the structural hypergraph and the molecular view, as well as between the semantic hypergraph and the molecular view, to enhance the representations learned from the molecular view. Finally, we conduct experiments on two real-world scientific datasets. The experimental results demonstrate a significant improvement of Mol-HCL over existing methods, showcasing its effectiveness and advantages in DDI prediction.

Index Terms—Drug-drug interaction, molecular-driven graph learning, multi-view hypergraph, contrastive learning.

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The data and source code are available at <https://github.com/ningq669/Mol-HCL>.

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I. INTRODUCTION

THE combination of multiple drugs can enhance the therapeutic effects of diseases to some extent, but it also increases the risk of adverse reactions caused by drug interactions [1]. Therefore, drug interaction detection becomes particularly important [2], [3]. Traditional methods for drug combination detection primarily rely on biological clinical experiments, but these approaches face issues such as high risks, long durations, low efficiency, and high costs [4], [5]. With advancements in science and technology, computational-based DDI identification has gained widespread attention [6], [7]. The existing methods for drug interaction identification can be categorized into three types: literature-based methods [8], similarity-based methods [9], and network-based methods [10].

Literature-based methods extract drug-drug interactions from biomedical texts [11], [12], [13]. However, literature-based methods often fail to fully capture the semantic features of drugs, resulting in incomplete drug information extraction and compromising the accuracy of DDI prediction.

Similarity-based methods predict potential drug-drug interactions by identifying known interacting drug pairs, finding drugs similar to one of the known drugs, and calculating drug similarity to infer possible DDI. Specifically, if a known interaction exists between drug i and drug j , and drug j is similar to drug k , it is inferred that drug i and drug k may interact. For instance, Zhang et al. [14] integrated various aspects of drug similarity, including chemical and biological similarities, and then employed ensemble learning for DDI prediction. Gottlieb et al. [15] assess the potential interactions between drug pairs for DDI prediction by calculating drug similarity, including chemical structure similarity, ligand similarity, and protein-protein interaction network similarity. Therefore, it is evident that similarity-based methods have drawbacks. They heavily rely on known DDIs, and struggle to capture the complex biological and chemical relationships between drugs, and have poor generalization capability.

Network-based methods refer to predicting drug-drug interactions by utilizing natural network structures, selecting specific network structures, or constructing custom network structures. Rohani et al. [16] combined and concatenated various known similarity information of drugs, and used neural networks for DDI prediction. Furthermore, Huang et al. [17] used an encoder to transform the functional representation of drug pairs into latent feature embeddings, which were then used to predict

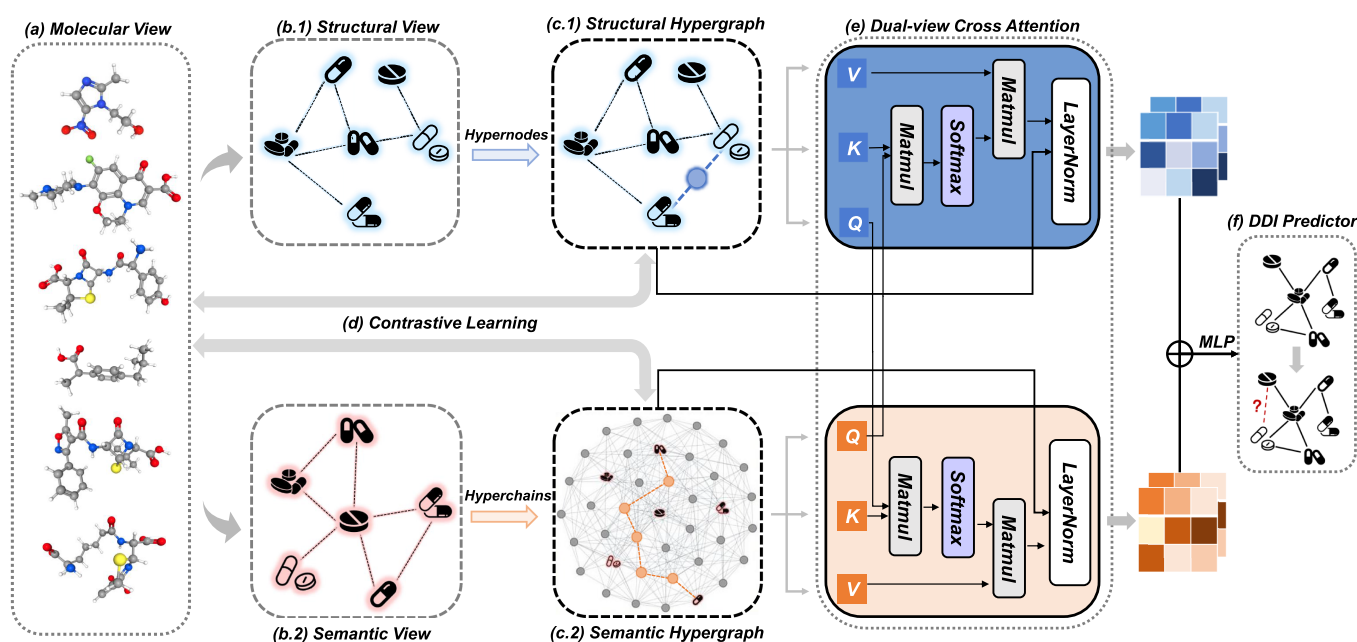


Fig. 1. Workflow of Mol-HCL: (a) Molecular View Learning; (b) Structural View and Semantic View Learning; (c) Structural Hypergraph and Semantic Hypergraph Learning; (d) Contrastive Learning; (e) Dual-views Cross Attention Learning; (f) DDI Predictor.

DDIs through a neural network. With the development of deep learning, methods applying graph learning to DDI prediction have gradually become prevalent [18], [19], [20], [21], [22], [23]. For example, Nyamabo et al. [24] employed a multi-layer Graph Attention Network (GAT) [25] to capture and aggregate the subspace information within drug molecules from multiple perspectives, creating drug molecule representations. However, single-view learning can only capture partial information of drug molecules and struggles to capture the multi-faceted features of drug molecules. Thus, multi-view contrastive learning methods that capture multi-level drug information were introduced. The inclusion of multiple views allows for a more in-depth and comprehensive learning of drug information [26], [27], [28], [29]. Zhang et al. [30] enhanced the focus on drug semantic information based on multi-view learning and introduced topological information, resulting in more enriched drug representations. Niu et al. [31] leverage Substructure Refined Representation Learning based on Self-Attention Mechanism to improve the performance of DDI prediction. Lea et al. [32] proposed a novel relational learning framework to predict interaction behavior between graph pairs by identifying core subgraphs within them. Zhang et al. [33] introduced the Iterative Substructure Extraction framework with Interactive Graph Information Bottleneck theory to precisely extract core interactive substructures and improve molecular relational learning. Li et al. [34] constructed two complementary views to enhance drug representations. By integrating molecular sequence features and multi-relational network topological features respectively, these views effectively mitigate the issues of data sparsity and class imbalance. Li et al. [35] addressed the challenge of fusing heterogeneous molecular information within biomedical knowledge graphs by integrating semantic embeddings generated by Generative Pre-trained Transformers (GPT) with Graph Neural Networks

(GNNs). This approach significantly enhances the model's predictive performance. Alawi et al. [36] integrated three main-stream backbone networks in parallel to capture rich and complementary multi-level features. By incorporating Multi-Scale Attention Module and Squeeze-Excite Refinement Block, the model dynamically reweights key features while suppressing background noise, thereby achieving efficient classification and diagnosis of colorectal diseases. Wang et al. [37] constructed deep feature representations of drugs from two distinct perspectives: molecular structure and knowledge graphs. By deeply integrating intra-molecular chemical structural information with external semantic associations from the knowledge graph, this model effectively addresses the limitation of information incompleteness inherent in single data sources. However, due to the current limitations of biotechnology, the shortcomings of traditional multi-view methods have resulted in a large number of hidden interactions between drugs that remain undiscovered.

To address these issues, we propose a hypergraph contrastive learning framework based on molecular view, as shown in Fig. 1. Building upon the molecular view, the structural and semantic views are first constructed. Hypernodes are then introduced into the structural view, and hyperchains (a chain composed of multiple hypernodes) into the semantic view, resulting in the formation of a structural hypergraph and a semantic hypergraph, respectively. It is worth noting that, in this study, structural information represents the direct neighbor interactions of drug nodes, while semantic information refers to the drug representations learned through long-range DDI chains. The construction of the structural hypergraph and semantic hypergraph enables us to establish pseudo one-hop connections, uncover hidden interaction information between drug nodes, and overcome the limitations imposed by multi-hop path propagation and topology-dependent information dissemination, thereby capturing high-order feature

information within the view. We then perform contrastive learning between the structural hypergraph, semantic hypergraph, and molecular view to enhance the performance and robustness of the model. The key innovations of this work are summarized as follows:

- The contrastive learning of the three-view framework (molecular view, structural hypergraph, and semantic hypergraph) extract intra-molecular and inter-molecular information from different perspectives. In the structural hypergraph, hypernodes model pharmacokinetic or pharmacodynamic relationships between drug pairs, enabling more connections between drug pairs to reveal latent DDIs. The integration of hyperchains allows the semantic view to overcome the limitations of multi-hop propagation and topology-dependent information dissemination, thereby facilitating the learning of higher-order drug features. Variations in relative positions between hypernodes and drug node pairs yield different feature proportions: proximal hypernodes favor structural features, while distant ones favor nonstructural information like pharmacological properties.
- The incorporation of cross-attention mechanisms enables mutual weighted fusion of features from structural hypergraph and semantic hypergraph, thereby enriching each hypergraph's original features with contextual information and relevant features from the other hypergraph.
- We designed three types of loss functions to enhance model accuracy and precision from different perspectives, including supervised loss, inconsistency loss, and contrastive loss, while simultaneously implementing three-view feature alignment.
- Comprehensive multi-perspective experimental analysis has fully validated the effectiveness of Mol-HCL in drug-drug interaction prediction.

II. METHODOLOGY

A. Model Architecture

1) *Molecular View*: The effects and properties of drugs are directly related to their chemical molecular structures. To investigate the molecular structures and drug properties of each drug, we construct molecular view. This involves extracting atomic and chemical bond features from drug molecules to form a feature matrix.

To represent each drug molecule more concisely and intuitively, we convert drug molecules into simplified molecular input line entry system (SMILES). Each SMILES encoding uniquely corresponds to a drug molecule, but a single drug molecule may have multiple SMILES representations. To avoid generating redundant or duplicate drug molecular view, we use RDKit [38] for processing. RDKit identifies the symbols in the SMILES strings and extracts the information on atoms and bonds, thereby constructing the drug molecular graph. In this graph, each atom is treated as a node, and each chemical bond is considered as an edge.

The atoms and chemical bonds of a drug molecule play a crucial role in its composition and properties. We leverage the

chemical bonds to transmit information between atoms, continuously updating the atomic representations. First, we extract the chemical bond features from the drug molecular graph, followed by linear transformations and activation function processing to obtain the initial hidden state information (message) for each bond. Subsequently, through multiple layers of DAMPN, the message aggregates the information of drug atoms, updating the atomic representations.

Specifically, the message initially propagates from each atom to its neighboring atoms. At each layer, the message aggregates information from multiple neighboring nodes and combines it with the atom's own features. This process iterates through multiple layers, updating the atomic representations and facilitating both message passing and the refinement of atomic information, as follows:

$$\tilde{a}_i^{(h)} = \sum_{c \in A_i} W_{ic}^{(h)} \cdot a_c^{(h-1)} \quad (1)$$

$$a_i^{(h)} = \sigma \left(W_a^{(h)} \cdot \left[a_i^{(h-1)}; \tilde{a}_i^{(h)} \right] \right) \quad (2)$$

where $\tilde{a}_i^{(h)}$ denotes the atomic embedding of atom node i at depth h after aggregating information from all its neighboring atoms, $a_i^{(h)}$ denotes the atomic embedding of atom i after aggregating its own features based on $\tilde{a}_i^{(h)}$, $\sum$ denotes the summation symbol, A_i denotes the set of directly neighboring atoms of atom i , including atom i itself, $W_{ic}^{(h)}$ denotes the matrix of trainable parameters that weights the messages received by node i from its neighboring node c at depth h , $a_c^{(h-1)}$ denotes the neighboring node c at depth $h-1$, $\sigma(\cdot)$ denotes the activation function, $[\cdot]$ denotes the concatenation operation of vectors.

After performing H iterations of propagation to form all the new hidden atomic representations, we use the attention mechanism to aggregate the atomic embeddings of all atoms within the given molecule, resulting in the atomic weighted combination representation t_{atom} , as follows:

$$t_{atom} = \sum_{i \in N_{atom}} \sigma \left(W_a \cdot \left[a_i^{(0)}; a_i^{(H)} \right] + b_a \right) \cdot a_i^{(H)} \quad (3)$$

where t_{atom} denotes the atomic weighted combination representation, W_a is a learnable weight matrix used to perform feature transformation on the input feature vectors, $a_i^{(0)}$ and $a_i^{(H)}$ represent the initial embedding of atom i and the hidden embedding vector at depth H , b_a denotes a bias term, N_{atom} represents the set of atoms within a given molecule.

Similarly, a linear transformation is applied to the chemical bond features, followed by an attention mechanism that weights and aggregates all chemical bond vectors within the drug molecule. Consequently, the combination of all chemical bond vectors is summarized as t_{bond} , as follows:

$$t_{bond} = \sum_{j \in N_{bond}} \sigma (W_b \cdot b_j + b_b) \cdot b_j \quad (4)$$

where N_{bond} denotes the set of bonds within the given molecule.

To more comprehensively reflect the structure and function of the drug molecule, we concatenate the atomic combination vector t_{atom} and chemical bond combination vector t_{bond} to

form a complete drug molecule representation t , as follows:

$$t = [t_{atom}; t_{bond}] \quad (5)$$

2) *Structural View and Semantic View Learning*: To better study the structure and properties of a drug molecule itself, we have established the molecular view. To more effectively learn the relational information between drug molecules, we build two inter-molecular views-structural view and semantic view-based on the molecular view to study the interactions between drug molecules. By leveraging these views, both the local and global contexts of molecular interactions can be captured, enhancing the understanding of the underlying mechanisms that govern their pharmacological effects.

In the inter-molecular view, each node represents a different drug molecule, and the edges connecting the nodes represent currently known DDI relationships. Based on the available DDI relationships between drug molecule pairs, an adjacency matrix can be derived to reveal the positional relationships between nodes in the network graph. Clearly, the positional relationships of drug nodes in the graph can be categorized into directly adjacent and non-directly adjacent nodes. This implies that for non-directly adjacent drug molecule pairs, their interaction relationships remain unknown.

Structural View: Based on currently known DDI relationships between drug node pairs, we employ Graph Attention Networks (GAT) to learn and aggregate feature information from one-hop neighboring nodes. By learning different aspects of information, such as molecular structures and pharmacological properties of neighboring nodes, we continuously enrich the feature representations of drug nodes, thereby facilitating the prediction of potential DDI relationships.

However, for a given drug node, it may interact with multiple one-hop neighboring nodes, and the impact of each interaction pattern on DDI varies. Additionally, the information triggering DDI reactions may originate from different drug properties, further increasing the complexity of prediction. Therefore, we introduce a multi-head attention mechanism to learn diverse feature representations of drug nodes, selectively highlighting important drugs while suppressing less relevant ones.

$$\gamma_{ij} = \text{softmax}(\text{attention}(t_i; t_j)) \quad (6)$$

$$E_{str_i} = \parallel_{s=1}^S \sigma \left(\sum_{j \in M_i} [\gamma_{ij}]_s \cdot t_j \right) \quad (7)$$

where γ_{ij} denotes the weight of drug j for drug i , $\text{softmax}(\cdot)$ denotes the normalization process, $\text{attention}(\cdot)$ indicates the attention score between node i and node j in the structural view, t_i and t_j represent the drug molecules i and j , E_{str_i} represents the structural drug embedding information of drug node i after being learned through the multi-head attention mechanism, $\parallel$ denotes the concatenation operation of multiple attention heads, $[\gamma_{ij}]_s$ represents the attention weight generated by the s -th attention head in the multi-head attention mechanism, used to weight the information flow between nodes i and j .

Semantic View: However, in practical clinical drug research, many newly developed drugs have limited or even no known DDI

relationships. The structural view, constrained by the characteristics of one-hop drug nodes, results in limited feature updates, making it inadequate for effectively predicting potential DDI associations. To address this limitation, we introduce semantic view, expanding the perspective from local drug pairs to a global drug network and from known DDI relationships to drug pairs that may contain potential DDI interactions.

To achieve this, we extend GAT to multiple layers and introduce a topology-aware coefficient β_{ij} , enabling the model to directly capture the n -hop topological information of nodes and learn non-adjacent drug interactions and relationships. Additionally, as the model depth increases, it can dynamically adjust the importance of different neighboring nodes.

$$\lambda_{ij}^{(r)} = \text{softmax} \left(\text{attention} \left(E_i^{(r-1)}; E_j^{(r-1)} \right) \right) \quad (8)$$

$$\eta_{ij}^{(r)} = \text{softmax} \left(w_\lambda \cdot \lambda_{ij}^{(r)} + w_\beta \cdot \beta_{ij} \right) \quad (9)$$

where $\lambda_{ij}^{(r)}$ represents the normalized attention weight of drug j for drug i at layer r (R in total), $E_i^{(r-1)}$ and $E_j^{(r-1)}$ represent the embedding information of drug i and drug j at layer $r-1$, respectively, $\eta_{ij}^{(r)}$ denotes the comprehensive attention score between drug node i and node j at layer r . w_λ and w_β are learnable parameters used to balance feature information and graph topological information, β_{ij} denotes the graph topological correlation between drug nodes i and j .

The resulting drug embedding representation is as follows:

$$E_{sem_i} = \parallel_{s=1}^S \sigma \left(\sum_{j \in M_i} [\eta_{ij}^{(R)}]_s \cdot E_j^{(R-1)} \right) \quad (10)$$

3) *Three-View Learning With Hypergraphs*. The addition of *Hypernodes and Hyperchains*: The incorporation of structural and semantic views enables a more comprehensive learning of different drug characteristics and DDI relationships. However, certain limitations still exist. The structural view is restricted to learning information from adjacent drug nodes, while the semantic view, despite capturing non-adjacent node information, remains constrained by model depth and the n -hop propagation range, leaving some potential DDI relationships unexplored.

To address these limitations, we introduce hypernodes into the structural view and hyperchains into the semantic view, and adopt a progressive learning strategy to enhance their performance. The structural view with hypernodes is referred to as the structural hypergraph, while the semantic view with hyperchains is termed the semantic hypergraph. In the structural view, the incorporation of hypernodes constructs pseudo one-hop neighbor relationships, directly linking previously non-adjacent drug node pairs. This significantly expands the scope of drug information that each node can learn, enabling the discovery of more potential drug interactions. In the semantic view, the incorporation of hyperchains effectively addresses the limitations of constrained iterative propagation ranges, establishes higher-quality node connections, and enriches the predictive modeling of drug interactions. Overall, the integration of hypernodes and hyperchains significantly enhances the accuracy and generalization capability of DDI prediction in the inter-molecular view.

Hypergraph Learning Process: After obtaining the drug embedding representations from the structural view learning, we use hypernodes to further learn and enrich the interaction relationships between drug nodes. First, we randomly initialize a set of vectors following a Gaussian distribution to serve as hypernodes. These vectors are dynamically inserted between pairwise drug representations generated by the GAT, constructing a unified node set comprising both drug nodes and hypernodes. Subsequently, we calculate the cosine similarity between nodes to obtain the connection strength between them, which represents their correlation, as follows:

$$R_{E_{str_i}, U_{str}} = \frac{E_{str_i} \cdot U_{str}}{\|E_{str_i}\|_2 \|U_{str}\|_2},$$

$$R_{E_{str_j}, U_{str}} = \frac{E_{str_j} \cdot U_{str}}{\|E_{str_j}\|_2 \|U_{str}\|_2} \quad (11)$$

$$R_{E_{sem_i}, U_{sem_k}} = \frac{E_{sem_i} \cdot U_{sem_k}}{\|E_{sem_i}\|_2 \|U_{sem_k}\|_2},$$

$$R_{E_{sem_j}, U_{sem_k}} = \frac{E_{sem_j} \cdot U_{sem_k}}{\|E_{sem_j}\|_2 \|U_{sem_k}\|_2} \quad (12)$$

where $R_{E_{str_i}, U_{str}}$ and $R_{E_{str_j}, U_{str}}$, $R_{E_{sem_i}, U_{sem_k}}$ and $R_{E_{sem_j}, U_{sem_k}}$ represent the cosine similarities between drug i , drug j , and the corresponding hypernode or hyperchain in the structural and semantic hypergraphs. E_{str_i} and E_{str_j} , E_{sem_i} and E_{sem_j} represent the drug feature representations of drug node i and node j in the structural and semantic views, respectively. U_{str} denotes a hypernode, U_{sem_k} denotes the smallest unit of a hyperchain, $\|\cdot\|_2$ denotes the L2 norm of a vector.

Subsequently, we use GCN to update the feature representations of the nodes:

$$H_{U_{str}}^{(l+1)} = \sigma \left(\tilde{G}_{U_{str}}^{-\frac{1}{2}} \delta \left(\tilde{B}_{U_{str}} \right) \tilde{G}_{U_{str}}^{-\frac{1}{2}} H_{U_{str}}^{(l)} W_{U_{str}}^{(l)} \right) \quad (13)$$

$$H_{U_{sem_k}}^{(l+1)} = \sigma \left(\tilde{G}_{U_{sem_k}}^{-\frac{1}{2}} \delta \left(\tilde{B}_{U_{sem_k}} \right) \tilde{G}_{U_{sem_k}}^{-\frac{1}{2}} H_{U_{sem_k}}^{(l)} W_{U_{sem_k}}^{(l)} \right) \quad (14)$$

where $H_{U_{str}}^{(l+1)}$ and $H_{U_{sem_k}}^{(l+1)}$ represents the structural and semantic embedding matrix of nodes at layer $l+1$, δ denotes Leaky ReLU activation function, $\tilde{G}_{U_{str}}^{-\frac{1}{2}}$ and $\tilde{G}_{U_{sem_k}}^{-\frac{1}{2}}$ denotes the normalized degree matrix, $\tilde{B}_{U_{str}}$ and $\tilde{B}_{U_{sem_k}}$ denotes the structural and semantic adjacency matrix with the identity matrix I added, composed of the cosine similarities between the three types of nodes. $H_{U_{str}}^{(l)}$ and $H_{U_{sem_k}}^{(l)}$ represents the structural and semantic embedding of the GCN at layer l , $W_{U_{str}}^{(l)}$ and $W_{U_{sem_k}}^{(l)}$ denotes the learnable weight matrix of structural and semantic hypergraphs

at layer l . The construction of $H_{U_{str}}^{(0)}$ and $H_{U_{sem}}^{(0)}$ is as follows:

$$H_{U_{str}}^{(0)} = \begin{pmatrix} E_{str_i} \\ E_{str_j} \\ U_{str} \end{pmatrix}, \quad H_{U_{sem}}^{(0)} = \begin{pmatrix} E_{sem_i} \\ E_{sem_j} \\ U_{sem_1} \\ U_{sem_2} \\ \vdots \\ U_{sem_n} \end{pmatrix} \quad (15)$$

It is clear that the drug information representations of drug i and drug j in the structural and semantic hypergraphs in each layer can be formulated as $E_{str_i}^{(l)}$ and $E_{str_j}^{(l)}$, $E_{sem_i}^{(l)}$ and $E_{sem_j}^{(l)}$.

After obtaining the drug node representations from each layer's GCN output, we use a hierarchical attention mechanism to obtain node representations with varying levels of importance. Then we compute the element-wise product of the embedding representations of two nodes to obtain the final drug feature representation after hypernode learning M_{str_i} and M_{sem_i} , as follows:

$$M_{str_i} = \text{CNN}(\text{Att}(E_{str_i}^q)) \odot \text{CNN}(\text{Att}(E_{str_j}^q)) \quad (16)$$

$$M_{sem_i} = \text{CNN}(\text{Att}(E_{sem_i}^q)) \odot \text{CNN}(\text{Att}(E_{sem_j}^q)) \quad (17)$$

where $\text{Att}(\cdot)$ represents the single-head attention mechanism, $\text{CNN}(\cdot)$ denotes the 1D convolutional neural network, $E_{str_i}^q$ and $E_{str_j}^q$, $E_{sem_i}^q$ and $E_{sem_j}^q$ represent the structural and semantic comprehensive representations of drug i and drug j after concatenating the multi-layer feature outputs, $\odot$ denotes the Hadamard product.

4) DDI Prediction. *DDI Prediction:* We compute the weighted sum of the structural view and the structural hypergraph, as well as the semantic view and the semantic hypergraph, as the final drug feature representations. This is equivalent to introducing residual connections, retaining some information before hypernode processing, which helps alleviate the vanishing gradient problem and promotes effective information flow, as follows:

$$f_{(i)} = \lambda M_{(i)} + (1 - \lambda) E_{(i)} \quad (18)$$

where λ is the residual weighting parameter for the structural (semantic) hypergraph, E_i and M_i represent the drug feature matrices for the structural (semantic) view and structural (semantic) hypergraph, respectively.

To better predict potential interactions between drugs, after the views learning for a pair of drug molecules, we use a fully connected neural network (FCN) to compress the two drug embeddings into an interaction link vector I_{ij} , representing the probability of interaction between drug i and drug j . Then, the sigmoid activation function is applied to generate the prediction scores I_{ij}^t and I_{ij}^f :

$$I_{ij}^t = \sigma[\text{FCN}(t_i \odot t_j)] \quad (19)$$

$$I_{ij}^f = \sigma[\text{FCN}(f_i \odot f_j)] \quad (20)$$

where $\text{FCN}(\cdot)$ represents FCN with two hidden layers, I_{ij}^f is the general term for the predicted scores of the structural hypergraph I_{ij}^{fstr} and semantic hypergraph I_{ij}^{fsem} .

To obtain the final drug representation for DDI prediction, the cross-attention mechanism is employed to process the drug feature matrices from the structural and semantic hypergraphs:

$$O_{str(i)}, O_{sem(i)} = \text{CrossAttention}(f_{str(i)}, f_{sem(i)}) \quad (21)$$

where $O_{str(i)}$ and $O_{sem(i)}$ represent the drug feature matrices from the structural hypergraph and semantic hypergraph after being processed by the cross-attention mechanism, $\text{CrossAttention}(\cdot)$ denotes the cross-attention mechanism.

Losses: The loss function consists of three components: supervision loss, inconsistency loss, and contrastive loss.

Supervision loss measures the distance between the predicted probability distribution and the actual labels, and is a commonly used loss function for binary classification problems. This loss not only measures the difference but also accelerates model convergence and enables online updates. We compute the difference between the final predicted scores of the three views and the true labels, summing them up:

$$\begin{aligned} \mathcal{L}_s = & \sum_{(i,j) \in \Phi} \text{BCE}(I_{ij}^t \parallel T_{ij}) + \text{BCE}(I_{ij}^{fstr} \parallel T_{ij}) \\ & + \text{BCE}(I_{ij}^{fsem} \parallel T_{ij}) \end{aligned} \quad (22)$$

where (i, j) represents a pair of drugs, Φ denotes the set of known drug pairs in the training dataset, T_{ij} denotes the true label for the drug pair (i, j) , $\text{BCE}(\cdot)$ denotes binary cross entropy loss function.

Inconsistency loss quantifies the dissimilarity between the three prediction results for unobserved samples. We calculate and minimize the inconsistency loss between any two views from the three views and sum them to improve the transparency and consistency between views:

$$\begin{aligned} \mathcal{L}_I = & \sum_{(i',j') \in \Phi^-} \text{KL}(I_{i'j'}^{fstr} \parallel I_{i'j'}^t) + \text{KL}(I_{i'j'}^{fsem} \parallel I_{i'j'}^t) \\ & + \text{KL}(I_{i'j'}^{fsem} \parallel I_{i'j'}^{fstr}) \end{aligned} \quad (23)$$

where (i', j') represents the unlabelled drug pair in the training set, and Φ^- denotes the set of unlabelled drug pairs in the training set, $\text{KL}(\cdot)$ denotes the Kullback–Leibler divergence.

To integrate and balance information from different views and maximize mutual information across views, contrastive learning is introduced and the contrastive loss is computed. Positive samples are defined as pairs formed between a drug’s local feature and the global features of both itself and all its one-hop neighbors. Conversely, negative samples consist of pairings between the target drug and all other drugs in the graph that lack direct connections.

We establish mutual information (MI) estimators for the structural view and semantic view to compute MI at both intra-molecular and inter-molecular levels. Specifically, we measure the MI between the structural hypergraph and the molecular view, as well as the MI between the semantic hypergraph and

the molecular view. However, in high-dimensional space, the distribution of drug data is highly complex, and the distances between data points are often large, making MI estimation challenging. Therefore, Jensen–Shannon Divergence (JSD) is introduced to measure the difference between two probability distributions. The use of JSD enables the model to learn the underlying structure and features of the drug data more effectively, thereby improving overall performance:

$$\begin{aligned} \hat{\mathcal{L}}_C = & \sum_{i \in \mathcal{M}} \frac{1}{|M_i|} \sum_{j \in M_i} \mathbb{Z}_{\mathbb{X} \times \mathbb{X}} [\text{sp}(\tau_{\mu, \rho}(t_j^\mu(x'); f_i^\rho(x)))] \\ & - \mathbb{Z}_{\mathbb{X}} [-\text{sp}(-\tau_{\mu, \rho}(t_j^\mu(x); f_i^\rho(x)))] \end{aligned} \quad (24)$$

where $\mathcal{M}$ and M_i represent the sets of all nodes and the set of drug node i itself and its directly adjacent drugs, $\mathbb{Z}_{\mathbb{X} \times \mathbb{X}}$ calculates the expected value of the joint samples under the two empirical probability distributions $\mathbb{X}$ and $\mathbb{X}$, and it evaluates the difference between the two distributions. $\text{sp}(\cdot)$ represents the softplus function, τ denotes feature transformation, μ represents the model parameters of the molecular view and ρ is for the structural (semantic) hypergraph similarly.

Therefore, we can obtain the $\hat{\mathcal{L}}_{C_1}$ of the structural hypergraph and $\hat{\mathcal{L}}_{C_2}$ of the semantic hypergraph. The final contrastive loss is the weighted sum of the losses produced by contrastive learning between the molecular view and the structural hypergraph, and between the molecular view and the semantic hypergraph:

$$\mathcal{L}_C = \hat{\mathcal{L}}_{C_1} + \hat{\mathcal{L}}_{C_2} \quad (25)$$

The total loss, denoted as $\mathcal{L}$, is obtained by the weighted combination of the three losses, as follows:

$$\mathcal{L} = \theta_1 \mathcal{L}_s + \theta_2 \mathcal{L}_I + \theta_3 \mathcal{L}_C \quad (26)$$

where θ_1 , θ_2 , and θ_3 represent the weighting parameters. To minimize the total loss, we optimize the model using the back-propagation algorithm.

III. EXPERIMENT

We evaluated our model on two publicly available datasets. Due to errors in the SMILES strings of some drugs in DrugBank, or because some entries were outdated and could not be properly converted into drug molecular graphs, we excluded these problematic data points. Specifically, the ZhangDDI dataset consists of 544 drugs and 45,720 drug-drug interaction (DDI) pairs, while the ChCh-Miner dataset includes 996 drugs and 21,486 DDI pairs. In these datasets, known DDIs between drugs are treated as positive samples, while unknown DDIs are treated as negative samples. For each dataset, we divide the data into training and testing sets in a 4:1 ratio and randomly select 25% of the training set as a validation set.

We evaluate the effectiveness of our model using four metrics: Area Under the Receiver Operating Characteristic Curve (AUROC), Average Precision (AP), F1 score (F1), and Accuracy (ACC). To reduce the impact of random errors, we repeat all experiments 10 times and calculate the average for the four metrics. In the comparison experiments, we compared our method

TABLE I
COMPARISON OF PERFORMANCE METRICS ACROSS DIFFERENT BASELINE METHODS

Dataset	Metric	MIRACLE	SSI-DDI	CSGNN	DeepDDS	DSN-DDI	HTCL-DDI	SRR-DDI	CGIB	IGIB-ISE	Mol-HCL
ZhangDDI	AUROC	96.44±0.35	93.14±0.29	91.71±0.09	93.20±0.23	91.13±0.15	98.20±0.41	82.82±0.72	94.91±0.04	95.32±0.21	98.74±0.05
	AP	93.09±0.53	92.09±0.39	89.02±0.73	92.08±0.31	86.42±0.30	96.32±0.46	60.19±3.13	92.76±0.12	94.10±0.14	97.05±0.09
	F1	85.16±0.27	81.96±1.24	83.60±0.73	82.79±0.42	87.68±0.40	89.93±0.91	53.14±2.08	84.66±0.05	85.54±0.58	91.75±0.87
	ACC	93.16±0.16	85.35±0.50	84.14±0.45	85.63±0.28	86.65±0.46	95.44±0.50	81.06±0.76	86.60±0.16	88.67±0.26	96.31±0.45
ChCh-Miner	AUROC	96.20±0.79	98.09±0.14	97.68±0.10	97.10±0.18	96.69±0.20	98.09±0.41	94.36±0.19	98.33±0.09	98.93±0.37	98.88±0.17
	AP	99.50±0.11	98.97±0.06	97.56±0.16	98.51±0.08	96.34±0.27	99.70±0.14	99.18±0.09	97.74±0.06	99.43±0.41	99.83±0.04
	F1	94.55±0.66	93.98±0.34	92.47±0.22	92.21±0.63	88.12±0.64	96.64±0.46	93.76±0.22	95.46±0.30	95.93±0.12	97.01±0.42
	ACC	90.77±0.11	92.19±0.48	92.54±0.17	90.38±0.64	88.89±0.42	94.21±0.37	89.36±0.21	94.24±0.27	94.83±0.42	94.83±0.70

with nine of the most advanced methods currently available, as detailed below:

A. Baselines

We compared our model with the following nine methods on two independent test sets:

- *MIRACLE* [28] uses multi-view contrastive learning to capture various features of drug molecules and DDIs.
- *SSI-DDI* [24] extracts features from drug molecular graphs and uses the substructural characteristics of drugs to transform the DDI prediction task between two drugs.
- *CSGNN* [27] employs a mixed-hop neighborhood aggregator to capture higher-order proximity in DDI networks and enhances transferability and generalization using contrastive graph neural networks.
- *DeepDDS* [20] utilizes Graph Convolutional Networks (GCNs) and attention mechanisms to compute drug embeddings, extracting information from both chemical structures and gene expression to identify specific synergistic drug combinations.
- *DSN-DDI* [29] is a dual-view learning network based on drug atomic-level information, learning drug substructures both within and between views.
- *HTCL-DDI* [30] uses three-view contrastive learning and introduces topological information, making drug information learning more comprehensive.
- *SRR-DDI* [31] is a DDI prediction model that leverages Substructure Refined Representation Learning based on a Self-Attention Mechanism.
- *CGIB* [32] is a novel relational learning framework that designed to predict interaction behavior between graph pairs by identifying core subgraphs within them.
- *IGIB-ISE* [33] is an Iterative Substructure Extraction framework with Interactive Graph Information Bottleneck theory, designed to iteratively extract core interactive substructures and enhance molecular relational learning.

B. Performance Comparison

Mol-HCL is compared with nine state-of-the-art methods on two independent test sets: ZhangDDI and ChCh-Miner. From the Table I, it is evident that the experimental results of our model outperform most other methods, demonstrating that multi-view contrastive learning and the inclusion of hypernodes

TABLE II
P-VALUES OF THE FOUR MODELS ON TWO DATASETS FROM TWO SIGNIFICANCE TESTS

Datasets	Tests	Metric	HTCL-DDI	SRR-DDI	CGIB	IGIB-ISE
ZhangDDI	Paired t-test	AUROC	1.29e-03	4.26e-15	4.90e-26	1.12e-14
		AP	9.16e-05	4.16e-07	3.56e-22	1.21e-18
		F1	9.24e-06	8.26e-13	1.04e-13	1.77e-12
		ACC	7.44e-14	5.56e-18	1.46e-16	2.86e-17
	Wilcoxon Signed-Rank Test	AUROC	9.77e-03	1.95e-03	1.95e-03	1.95e-03
		AP	1.95e-03	1.95e-03	1.95e-03	1.95e-03
		F1	1.95e-03	1.95e-03	1.95e-03	1.95e-03
		ACC	1.95e-03	1.95e-03	1.95e-03	1.95e-03
ChCh-Miner	Paired t-test	AUROC	1.15e-07	2.54e-20	4.74e-10	6.54e-02
		AP	8.96e-04	1.11e-12	6.06e-21	2.18e-02
		F1	5.05e-08	8.56e-14	6.33e-13	2.65e-08
		ACC	3.06e-06	2.23e-12	2.30e-05	7.60e-02
	Wilcoxon Signed-Rank Test	AUROC	1.95e-03	1.95e-03	1.95e-03	1.31e-01
		AP	3.91e-03	1.95e-03	1.95e-03	3.71e-03
		F1	1.95e-03	1.95e-03	1.95e-03	1.95e-03
		ACC	1.95e-03	1.95e-03	1.95e-03	4.32e-01

significantly enhance DDI prediction performance. The contrastive learning between intra-molecular and inter-molecular views facilitates mutual information maximization across different perspectives, enabling consistent representation of identical nodes across multi-angle views. The integration of hypernodes and hyperchains further empowers the model to capture latent DDIs while expanding its capacity to explore neighbor drug node structural relationships and long-range DDI chain semantic information, thereby enhancing the discovery and prediction of broader-spectrum DDIs. In Table S1 (in the Appendix), we calculate and report the number of parameters, average training time per epoch, as well as the total training and inference times for our model and 9 baseline methods across two datasets. This comparison is intended to demonstrate the performance and efficiency of our proposed model.

We conducted the Paired t-test and the Wilcoxon Signed-Rank Test to validate that our proposed model achieves significant improvements over state-of-the-art baselines from the last two years on four metrics. In Table II, the vast majority of p-values are less than 0.05, which demonstrates the statistical significance of our model's performance improvement.

C. Ablation Study

The characteristic of Mol-HCL lies in the introduction of contrastive learning across three views, introduction of the unsupervised losses, and the addition of hypernodes to both the structural view and semantic view. To explore the contribution

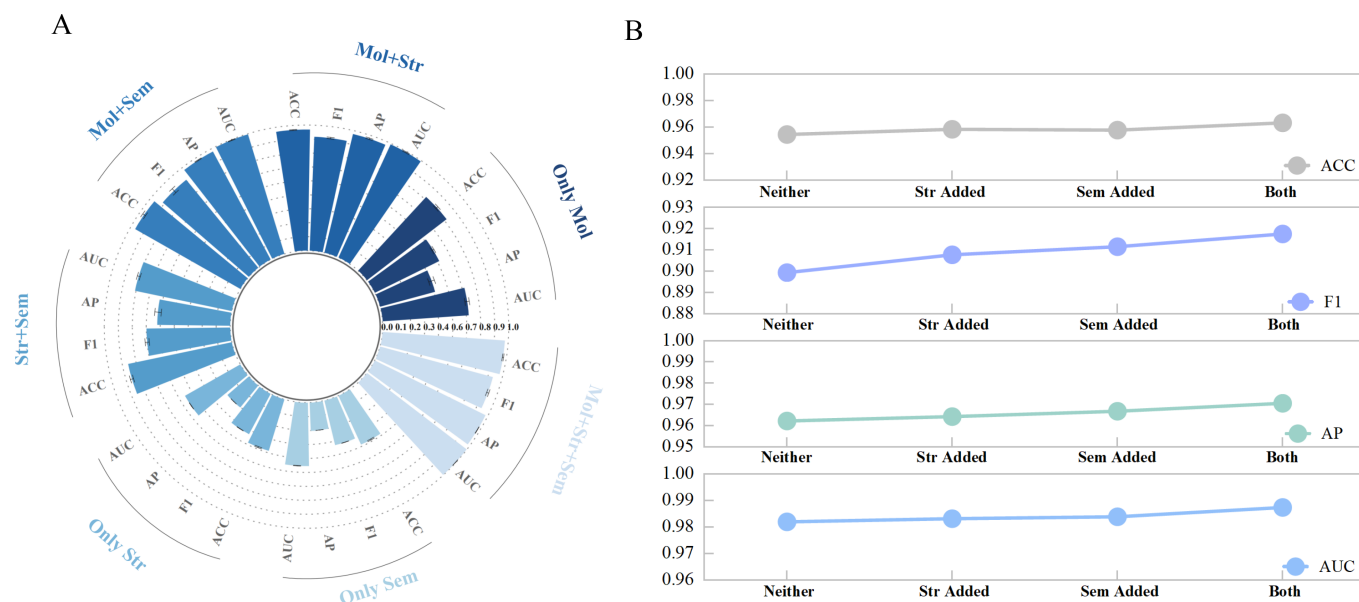


Fig. 2. The impact of (a) the number of views, (b) the inclusion of hypernodes and hyperchains on the experimental results on the ZhangDDI dataset.

of the above structure to our method, we conducted four ablation experiments.

1) *Analysis of the Impact of Different Views*: To investigate the impact of view quantity on model performance, experiments were conducted with varying numbers of views. In Fig. 2 (A), for example, “Mol” refers to the molecular view only, “Mol+Str” adds the structural hypergraph, and “Mol+Sem+Str” combines all three views. The results show that the molecular view alone yields unsatisfactory performance. However, adding the structural or semantic hypergraph significantly improves performance, with the best results achieved by combining all three views. Mol-HCL outperforms the variants in four evaluation metrics. Compared to the optimal values of the other six conditions, it achieves an improvement of 0.31% in AUROC (compared to Mol+Sem), 0.51% in AP (compared to Mol+Sem), 2.08% in F1-score (compared to Mol+Str), and 0.90% in ACC (compared to Mol+Str). These results indicate that the incorporation of three views allows for a more complete learning of drug information, compensating for the limitation of the molecular view, which can only learn internal drug information, and the single limitation of the dual-view learning approach, making DDI prediction more accurate. In Table S2 (in the Appendix), the detailed numerical results are presented.

2) *Analysis of the Impact of Joining the Hypernodes and Hyperchains*: To explore the impact of adding hypernodes and hyperchains into the views, we conducted experiments. In Fig. 2 (B), “Neither” means neither view has hypernodes or hyperchains. “Str Added” refers to the structural view with hypernodes, “Sem Added” to the semantic view with hyperchains, and “Both” to structural view with hypernodes and semantic view with hyperchains. As the Fig. 2(b) shows, the average AUROC of Mol-HCL is 0.9874, surpassing that of “Str Added” (0.9820), “Sem Added” (0.9829), and “Neither” (0.9809). Similarly, Mol-HCL achieves an average AP of 0.9705, outperforming “Str Added” (0.9631), “Sem Added” (0.9648), and “Neither”

(0.9613). In addition, the average F1 score and ACC of Mol-HCL exceed the best results among all other methods by 0.87% and 0.39%, respectively. Experimental results demonstrate that incorporating hypernodes into the structural view and introducing hyperchains of specific lengths into the semantic view effectively enhance the model’s ability to capture structural and semantic information between drug nodes, thereby achieving improved DDI prediction performance. In Table S3 (in the Appendix), the detailed numerical results are presented.

3) *Analysis of the Impact of Various Losses*: To investigate the impact of different loss functions on model performance, a series of experiments was conducted. Fig. 3 shows the impact of two unsupervised losses on DDI prediction. “No DGILoss” indicates the presence of only supervision and inconsistency losses, while “No KLLoss” means only supervision and contrastive losses are included. “Exist Both” represents the inclusion of all three losses. We removed either the inconsistency loss or the contrastive loss, keeping the supervision loss, to evaluate their contributions.

The average AUROC of Mol-HCL is 0.9874, which exceeds the values of “No DGILoss” (0.9831) and “No KLLoss” (0.9847). In terms of average AP, Mol-HCL also leads with a score of 0.9705, surpassing “No DGILoss” (0.9689) and “No KLLoss” (0.9687). Moreover, Mol-HCL outperforms the best results of other methods by 0.60% in average F1 score and 0.30% in ACC. The results demonstrate that adding unsupervised losses improves DDI prediction performance: the inconsistency loss enhances model robustness by reducing prediction differences across views, and the contrastive loss boosts the model’s ability to distinguish sample relationships by adjusting samples’ feature space distances. In Table S4 (in the Appendix), the detailed numerical results are presented.

4) *Analysis of the Distinct Feature Fusion Methods*: To explore the impact of different feature fusion methods between the structural and semantic views on experimental results, we

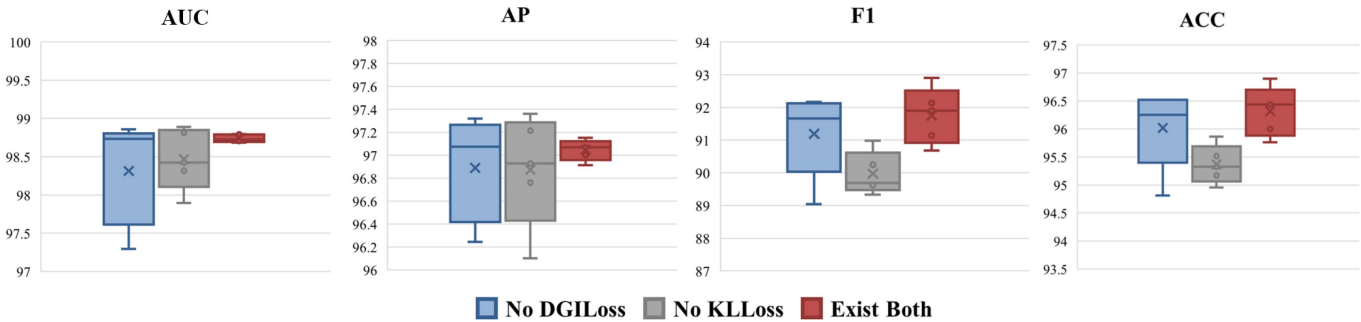


Fig. 3. The impact of the kind of losses on the experimental results on the ZhangDDI dataset.

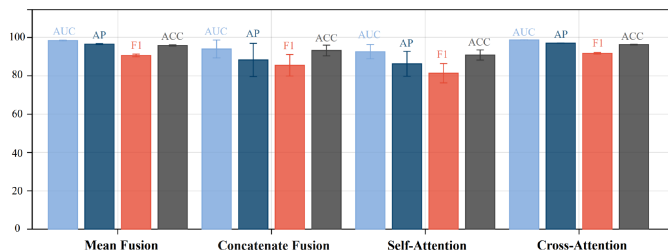


Fig. 4. The impact of distinct feature fusion methods on experimental results on the ZhangDDI dataset.

conducted a series of experiments. As shown in Fig. 4, “Mean Fusion” refers to summing the feature matrices of the structural and semantic views and then dividing the result by two, “Concatenate Fusion” denotes concatenating the two feature matrices, “Self-Attention” represents processing both feature matrices using the self-attention mechanism, and “Cross-Attention” applies the cross-attention mechanism to them.

Mol-HCL achieved better performance over all other experimental settings across the four evaluation metrics. Relative to the best results among the other three alternative configurations, it attained an AUROC improvement of 0.30% and an AP improvement of 0.48%, as well as 1.2% in F1-score and 0.51% in ACC. The experimental results indicate that the cross-attention mechanism achieves the best performance. This is because computing the average and concatenation operations merely combine the features of the two hypergraphs without explicitly capturing their interrelations, and self-attention focuses only on intra-feature dependencies. In contrast, cross-attention allows the query of one hypergraph to attend to the key and value of the other, enabling both hypergraphs to extract the most relevant feature information from each other. This mechanism effectively models the complex interactions between the two hypergraphs. In Table S5 (in the Appendix), the detailed numerical results are presented. We have plotted the heatmaps of the cross-attention weight matrices for the two hypergraphs and analyzed the information exchange between them. Please refer to the Appendix for details.

5) *Analysis of the Impact of JSD*: To investigate the significance of JSD, we conducted a series of experiments. Table III illustrates the impact of five distinct loss functions on model performance within the context of contrastive learning. Notably, the model employing JSD as the graph contrastive

TABLE III
THE IMPACT OF JSD ON THE EXPERIMENTAL RESULTS ON THE ZHANGDDI DATASET

Loss	AUROC	AP	F1	ACC
InfoNCE	84.58±0.24	85.46±0.08	73.81±0.10	76.92±0.01
Circle Loss	96.24±4.42	93.50±6.21	90.43±1.51	95.77±0.75
Sigmoid Loss	98.57±0.11	96.76±0.01	88.95±1.42	94.82±0.81
PolyLoss	83.49±1.90	82.51±5.19	73.82±0.41	77.02±0.27
JSD	98.74±0.05	97.05±0.09	91.75±0.87	96.31±0.45

learning objective achieves superior performance across all four metrics. Although InfoNCE represents a prevailing paradigm in contrastive learning, it treats all unobserved DDIs as negative samples and repels them in the feature space. This leads to the complete neglect of potential DDIs, thereby hindering the model’s predictive effectiveness. Circle Loss sharpens the decision boundary by dynamically adjusting sample weights; however, in the presence of noisy DDI data, it tends to overfit label noise, consequently degrading the model’s generalization capability. Sigmoid Loss processes each edge in isolation, severing the dependencies between nodes, their neighborhoods, and the global graph topology, which limits the model to learning only local features. While Poly Loss effectively mitigates data imbalance by reshaping the gradient distribution, it inherently fails to incorporate graph structural information. information between local node representations and global graph summaries, JSD enables the model to effectively capture structural information within the drug network. Furthermore, its formulation based on statistical expectation renders the model significantly more robust to noise.

D. Hyperparameters Analysis

To evaluate the robustness of Mol-HCL with respect to key architectural and optimization hyperparameters, we conducted sensitivity analyses on three critical factors: the residual weights for the structural and semantic hypergraphs, the number of hypernodes in both hypergraphs, and the weighting coefficients of the three losses. The experimental results show that Mol-HCL consistently achieves high performance across a wide range of parameter settings. For example, AUROC remains above 0.9718 for all tested residual weights, reaching a peak of 0.9874. When the number of hypernodes is varied from 1 to 8 in the structural hypergraph and from 8 to 64 in the semantic hypergraph, only

minor fluctuations are observed (less than 1% in AUROC and AP), indicating structural stability. Similarly, the model demonstrates strong resilience to changes in the loss weight combinations, with optimal performance achieved when slightly more emphasis is placed on the supervised and contrastive losses. These results suggest that Mol-HCL is robust to hyperparameter variations and can maintain reliable performance across diverse configuration.

1) *Analysis of the Impact of the Residual Weights:* Introducing residual connections in the structural hypergraph and semantic hypergraph mitigates gradient vanishing or explosion problems, prevents model degradation, and improves performance and generalization. To explore the impact of residual weights, we conducted experiments on the residual weights of the structural hypergraph and semantic hypergraph. Fig. S2 (in the Appendix) shows the effect of varying the residual parameter. The range of 0.3 to 0.7 represents the residual weight range, chosen after multiple experiments to prevent the model from overly relying on one input. Results show that when $\lambda = 0.5$, the structural hypergraph's performance is optimal, and when $\lambda = 0.6$, the semantic hypergraph's performance is optimal. The fact that the values of λ are all centered indicates that both explicit and hidden information are equally important for DDI prediction. In Table S5 (in the Appendix), the detailed numerical results are presented.

2) *Analysis of the Number of Hypernodes and the Length of Hyperchains:* To explore the impact of hypernode numbers and the length of hyperchains on the experimental results, a series of experiments was conducted. Fig. S3 (in the Appendix) shows the impact of varying the number of hypernodes and the length of hyperchains on the experimental results. When the number of hypernodes in the structural hypergraph is set to 1 and the length of hyperchains in the semantic hypergraph is set to 32, optimal experimental results are achieved. The structural hypergraph aims to acquire and learn both known and latent one-hop node information of drug nodes, while continuously updating drug node representations. In the structural hypergraph, when the number of hypernodes between two drug nodes becomes excessive, the significance of originally known one-hop information undergoes progressive dilution during information propagation through hypernode chains. This phenomenon ultimately impedes the structural hypergraph's capacity to capture hidden DDI information. Experimental results demonstrate that the performance of the structural hypergraph is optimal when the number of hypernodes is set to 1. The purpose of the semantic hypergraph is to acquire and learn both known and potential n-hop drug chain information, which often brings rich and implicit semantic representations of long-range drug interactions. In the semantic hypergraph, the integration of multi-layer GAT and topological coefficients empowers it to overcome the inherent structural constraints of DDI networks. This configuration supports the progressive prediction of non-adjacent node interactions, thereby allowing hyperchains of a certain length to exist between drug nodes while maintaining effective information propagation. At the same time, the length of the hyperchain is also crucial. Hyperchains that are too short may result in sparse semantic information between drug pairs, while

excessively long hyperchains could introduce noise from unnecessary node propagation. Through our ablation experiments, we found that the model performs optimally when the hyperchain length is set to 32. The aforementioned theory thoroughly validates the rationality of the experimental results. In Table S7 (in the Appendix), the detailed numerical results are presented.

3) *Analysis of the Impact of Loss Weights:* To explore the impact of the three losses on the experimental results, we use loss weights to determine the importance of each loss to the experimental outcomes. We experimented with setting the weights of the three losses to 0.8, 1, and 1.2, and chose AUC and AP as the primary performance evaluation metrics to plot the graph. In Fig. S4 (in the Appendix), node size is used to measure the AUC, and the shading of the node color is used to measure the AP value. It is evident that all three losses are important, but when $\theta_1 = 1$, $\theta_2 = 0.8$, and $\theta_3 = 1$, the node size is the largest and the color is the most vibrant indicating the best experimental results under this set of parameters. The weights of the supervised loss and contrastive loss are both set to 1, indicating their dominant role in the model training process. This allows the model to continuously fit the labels while also learning highly discriminative drug representations, thereby improving prediction accuracy and reducing the risk of overfitting. The weight of the inconsistency loss is set to 0.8, slightly lower than the other two, suggesting that the model encourages a degree of divergence between the three views to facilitate mutual complementarity among them.

In Table S8, we detail the model's performance, in terms of AUROC and AP, on two datasets across varying configurations of the three weight parameters. Typically, sparse networks render model training highly sensitive to hyperparameters due to the scarcity of structural information. However, our experimental results demonstrate that, despite the Miner dataset being significantly sparser than the Zhang dataset, the model achieved optimal performance across both metrics under the identical parameter configuration ($\theta_1 = 1$, $\theta_2 = 0.8$, $\theta_3 = 1$). This finding indicates that our model exhibits exceptional robustness, independent of the specific density distribution of the datasets. Consequently, in practical applications, our model can be generalized to DDI networks with varying sparsity levels without the need for extensive dataset-specific fine-tuning.

E. Case Study

To further investigate the reliability of the prediction results of Mol-HCL, we selected the top 20 drug pairs with the highest predicted scores related to Metronidazole from the ZhangDDI test set. These pairs were analyzed in conjunction with drug interaction records from the DrugBank database, other published papers, and model predictions.

In Table S9 (in the Appendix), the drug interaction evidence for the fifth drug pair comes from the paper [39], which documented the treatment of systemic tetanus in kittens. The combined use of Methocarbamol and Metronidazole led to the cat's gradual deterioration in health. Methocarbamol, a CNS depressant, and Metronidazole, which can

cause neurological side effects at high doses, may have additive CNS effects when co-administered. This increases the risk of drowsiness, dizziness, cognitive impairment, and excessive sedation, especially in elderly patients or those with liver dysfunction.

The drug interaction evidence for the eighteenth drug pair is based on paper [40]. Experiments have shown that the combination of Metronidazol amino acidum natricum and Carboplatin significantly inhibits tumor growth. Since Metronidazol amino acidum natricum is a water-soluble sodium salt derivative of Metronidazole, which releases Metronidazole in the body and exerts the same antibacterial effects, it can be inferred that a drug-drug interaction also exists between Metronidazole and Carboplatin.

For detailed inferences regarding the relationships of the unknown DDI pairs among the top-20 predictions for Metronidazole, please refer to the Appendix.

IV. CONCLUSION

In this paper, we explore both intra-molecular information and inter-molecular interactions. To this end, we construct a hypergraph contrastive learning framework based on molecular view, including molecular view, structural hypergraph, and semantic hypergraph, with the incorporation of hypernodes in the structural hypergraph and hyperchains in the semantic hypergraph. The molecular view is used to investigate the structure and properties of drug molecules, while the structural hypergraph and semantic hypergraph calculate the interaction information between drug molecules from the perspectives of one-hop structural information and n-hop semantic information, respectively. The incorporation of hypernodes into the structural view enables the construction of pseudo one-hop neighbor nodes, allowing the model to uncover more latent DDI information. Meanwhile, the introduction of hyperchains in the semantic view breaks through the limitations of n-hop propagation, facilitating the capture and learning of information from distant drug nodes and enabling the computation of more reliable prediction scores. As a result, both the efficiency and accuracy of DDI prediction are significantly improved. The introduction of contrastive learning enhances the model's ability to distinguish between negative and positive drug pairs, boosting both its performance and robustness. We conducted experiments on two public datasets, and the results demonstrate that this method provides strong support for drug interaction prediction. Our work can be further improved in the following ways: (1) integrating multimodal data to enhance the model's comprehensiveness and accuracy in drug interaction prediction; (2) incorporating multi-task learning to share potential features across models, improving overall prediction performance; (3) incorporating a variational Bayesian learning mechanism to prune noisy connections within the drug network and integrate the generated view for contrastive learning [41], while simultaneously leveraging self-attention to capture long-range dependencies and mitigate the over-smoothing issue caused by stacking multiple GAT layers [42]; (4) applying our model to more practical use cases.

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